

Advent Of Code Solutions

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Chapter 1

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```
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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 4

File Index

4.1 File List

Here is a list of all documented files with brief descriptions:

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Chapter 5

Class Documentation

5.1 `histogram_lock< max_elem, min_elem, T >` Class Template Reference

Public Member Functions

- `histogram_lock` (T init)
Construct a new lock object at position init.
- void `setCurrentValue` (T val)
sets the current position of the lock
- void `operator==` (T val)
Sets the current position of the lock.
- void `operator+` (T val)
Increases the position of the lock by val amount.
- void `operator-` (T val)
- const T `get_current_value` () const
- T `get_frequency` (T val)
Gets how many times the lock was at val.

5.1.1 Detailed Description

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
class histogram_lock< max_elem, min_elem, T >
```

Definition at line 36 of file [lock.hpp](#).

5.1.2 Constructor & Destructor Documentation

5.1.2.1 `histogram_lock()`

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
histogram_lock< max_elem, min_elem, T >::histogram_lock (
    T init ) [inline]
```

Construct a new lock object at position init.

Parameters

<i>init</i>	
-------------	--

Definition at line 43 of file [lock.hpp](#).

5.1.3 Member Function Documentation

5.1.3.1 `get_current_value()`

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
const T histogram_lock< max_elem, min_elem, T >::get_current_value ( ) const [inline]
```

Definition at line 74 of file [lock.hpp](#).

5.1.3.2 `get_frequency()`

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
T histogram_lock< max_elem, min_elem, T >::get_frequency (
    T val ) [inline]
```

Gets how many times the lock was at val.

Parameters

<i>val</i>	
------------	--

Returns

count

Definition at line 81 of file [lock.hpp](#).

5.1.3.3 `operator+()`

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
void histogram_lock< max_elem, min_elem, T >::operator+ (
    T val ) [inline]
```

Increases the position of the lock by val amount.

Parameters

<i>val</i>	
------------	--

Definition at line 68 of file [lock.hpp](#).

5.1.3.4 operator-()

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
void histogram_lock< max_elem, min_elem, T >::operator- (
    T val ) [inline]
```

Definition at line 72 of file [lock.hpp](#).

5.1.3.5 operator==()

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
void histogram_lock< max_elem, min_elem, T >::operator== (
    T val ) [inline]
```

Sets the current position of the lock.

Parameters

<i>val</i>	
------------	--

Definition at line 61 of file [lock.hpp](#).

5.1.3.6 setCurrentValue()

```
template<std::size_t max_elem, std::size_t min_elem = 0, std::signed_integral T = std::int32_t>
void histogram_lock< max_elem, min_elem, T >::setCurrentValue (
    T val ) [inline]
```

sets the current position of the lock

Parameters

<i>val</i>	
------------	--

Definition at line 50 of file [lock.hpp](#).

The documentation for this class was generated from the following file:

- 2025/Day1/[lock.hpp](#)

Chapter 6

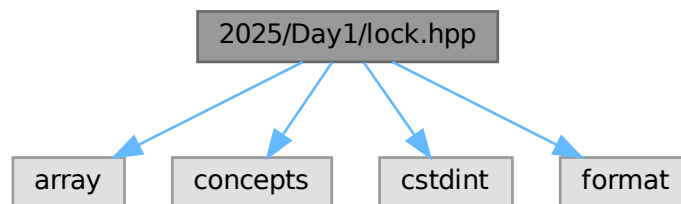
File Documentation

6.1 2025/Day1/lock.hpp File Reference

Implements a lock that holds a histogram of values.

```
#include <array>
#include <concepts>
#include <cstdint>
#include <format>
```

Include dependency graph for lock.hpp:



Classes

- class [histogram_lock](#)< max_elem, min_elem, T >

6.1.1 Detailed Description

Implements a lock that holds a histogram of values.

Author

Dylan Shah (code@dylan-shah.com)

Version

0.1

Date

2026-05-13

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Definition in file [lock.hpp](#).

6.2 lock.hpp

[Go to the documentation of this file.](#)

```
00001 /**
00002  * \file lock.hpp
00003  * \author Dylan Shah (code@dylan-shah.com)
00004  * \brief Implements a lock that holds a histogram of values.
00005  * \version 0.1
00006  * \date 2026-05-13
00007  *
00008  * \copyright Copyright (c) 2026
00009  *
00010  * \parblock License:
00011  *
00012  * This program is free software: you can redistribute it and/or modify
00013  * it under the terms of the GNU General Public License as published by
00014  * the Free Software Foundation, either version 3 of the License, or
00015  * (at your option) any later version.
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00017  * This program is distributed in the hope that it will be useful,
```

```

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00019 *   MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
00020 *   GNU General Public License for more details.
00021 *
00022 *   You should have received a copy of the GNU General Public License
00023 *   along with this program.  If not, see <https://www.gnu.org/licenses/>.
00024 * \endparblock
00025 *
00026 */
00027
00028 #include <array>
00029 #include <concepts>
00030 #include <cstdint>
00031 #include <format>
00032
00033 template<std::size_t          max_elem,
00034         std::size_t          min_elem = 0,
00035         std::signed_integral T      = std::int32_t>
00036 class histogram_lock {
00037 public:
00038     /**
00039     * \brief Construct a new lock object at position init
00040     *
00041     * \param init
00042     */
00043     histogram_lock(T init) { setCurrentValue(init); }
00044
00045     /**
00046     * \brief sets the current position of the lock
00047     *
00048     * \param val
00049     */
00050     void setCurrentValue(T val) {
00051         bounds_check(val);
00052         current_value = std::move(val);
00053         histogram[val - min_elem]++;
00054     }
00055
00056     /**
00057     * \brief Sets the current position of the lock
00058     *
00059     * \param val
00060     */
00061     void operator==(T val) { setCurrentValue(val); }
00062
00063     /**
00064     * \brief Increases the position of the lock by \ref val "val" amount
00065     *
00066     * \param val
00067     */
00068     void operator+(T val) {
00069         setCurrentValue((current_value + val) % range + min_elem);
00070     }
00071
00072     void operator-(T val) { this->operator+(-1 * val); }
00073
00074     const T get_current_value() const {return current_value;}
00075     /**
00076     * \brief Gets how many times the lock was at val
00077     *
00078     * \param val
00079     * \return count
00080     */
00081     T get_frequency(T val) {
00082         bounds_check(val);
00083         return histogram[val - min_elem];
00084     }
00085
00086 private:
00087     std::array<T, max_elem - min_elem> histogram{};
00088     T current_value;
00089
00090     static constexpr T range = max_elem - min_elem;
00091
00092     /**
00093     * \brief Checks if the argument is in the range that this lock can handle
00094     *
00095     * \param t checked value
00096     * \return Throws std::range_error if out of range or returns true
00097     */
00098     bool bounds_check(T t) {

```

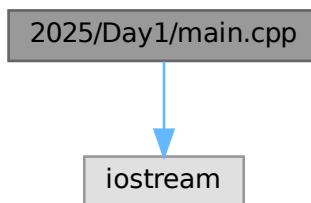
```
00099         if (t > max_elem || t < min_elem)
00100             throw std::range_error{ std::format(
00101                 "Value {} not in lock bounds [{} - {}]", t, min_elem,
00102                 max_elem) };
00103         return true;
00104     }
00105 };
```

6.3 2025/Day1/main.cpp File Reference

My Day One Solution Problem.

```
#include <iostream>
```

Include dependency graph for main.cpp:



Functions

- int [main](#) (int argc, char const *argv[])

6.3.1 Detailed Description

My Day One Solution Problem.

Author

Dylan Shah (code@dylan-shah.com)

Version

0.1

Date

2026-05-13

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Definition in file [main.cpp](#).

6.3.2 Function Documentation

6.3.2.1 main()

```
int main (
    int argc,
    char const * argv[] )
```

Definition at line 31 of file [main.cpp](#).

6.4 main.cpp

[Go to the documentation of this file.](#)

```
00001 /**
00002  * \file main.cpp
00003  * \author Dylan Shah (code@dylan-shah.com)
00004  * \brief My Day One Solution
00005  * \ref https://adventofcode.com/2025/day/1 "Problem"
00006  * \version 0.1
00007  * \date 2026-05-13
00008  *
00009  * \copyright Copyright (c) 2026
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00022  *
00023  * You should have received a copy of the GNU General Public License
00024  * along with this program. If not, see <https://www.gnu.org/licenses/>.
00025  * \endparblock
00026  *
00027  */
00028
00029 #include <iostream>
00030
00031 int main(int argc, char const* argv[]) {
00032     std::cout << "test";
00033     return 0;
00034 }
```


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